Macromolecular Research, Vol. 19, No. 2, pp 203-208 (2011)
DOI 10.1007/s13233-011-0216-2 www.springer.com/13233

Preparation and Characterization of Nanocomposite Based on Polyaniline and Graphene Nanosheets

Ngo Trinh Tung[1,3], Tran Van Khai[2], Minhee Jeon[1], Yeo Jin Lee[1], Hoeil Chung[1], Jeong-Hwan Bang[4], and Daewon Sohn*[,1]

[1]*Department of Chemistry, Hanyang University, Seoul 133-791, Korea*
[2]*Division of Advanced Materials Science Engineering, Hanyang University, Seoul 133-791, Korea*
[3]*Institute of Chemistry, Vietnam Academy of Science and Technology (VAST), 18 Hoang Quoc Viet, Caugiay, Hanoi, Vietnam*
[4]*Department of Environmental Health, Seonam University, Namwon 590-711, Korea*

Received September 17, 2010; Revised October 20, 2010; Accepted October 30, 2010

Abstract: Polymer nanocomposites based on polyaniline (PANi) and graphene nanosheets (GNS) modified with poly(sodium 4-styrensulfonate) (PSS-GNS) were prepared, and their structure and properties were investigated by atomic force microscopy (AFM), scanning electron microscopy (SEM), UV-vis spectroscopy, ATR-IR spectroscopy, X-ray diffraction, elemental analysis, thermogravimetric analysis (TGA) and electrical conductivity measurements. The results revealed that for the PANi/PSS-GNS nanocomposites, the disordered structure of PSS-GNS was fully destroyed and PSS-GNS exists in the form of a single GNS or stacked PSS-GNS elements in a PANi matrix. PSS-GNS was partly covered by PANi due to hydrogen bonding that occurs between the PSS-GNS and PANi. By incorporating PSS-GNS, the electrical conductivity of PANi increased linearly from 0.84 S/cm for neat PANi to 4.96 S/cm for a PANi/PSS-GNS (5%) nanocomposite. The thermal stability of the PANi was also improved significantly to approximately 100 °C by the nanocomposite.

Keywords: polyaniline, graphene nanosheet, conductivity, poly(sodium 4-styrensulfonate).

Introduction

Intrinsically conducting polymers (ICPs) are promising materials for applications in electronics, electrochemistry, electroluminescence, membranes and sensors.[1,2] Among the ICPs, polyaniline (PANi) has attracted considerable attention because of its easy synthesis, low monomer cost, good environmental and thermal stability as well as adequate electrical conductivity.[3,4] To improve the properties of PANi, especially the electrical conductivity, several conductive nanofillers including gold nanoparticles,[5,6] silver nanoparticles[7] or carbon nanotubes[8] (CNTs) have been used for preparation of PANi nanocomposites.

In recent years, graphene nanosheets (GNSs) consisting of one to a few graphene layers have been recognized as a promising, cost effective and high quality alternative to CNTs and carbon nanofibers in composite applications. GNSs exhibit outstanding electrical and thermal conductivity comparable to CNTs, while the specific surface area is twice that of CNTs.[9,10]

GNS are mostly prepared by physical methods such as micro-mechanical cleavage of graphite[11] or by chemical reduction of exfoliated graphene oxide nanosheets.[12,13] Among these methods, preparation of GNS from exfoliated graphene oxide nanosheets has attracted much research interest because this method is efficient and results in high yields of single-layered graphene oxide. The main drawback of this method is that the graphene oxide nanosheets contain several oxygen functional groups, including hydroxyl, epoxy, carbonyl and carboxyl groups, that make graphite oxide nanosheets strong hydrophilic characters and electrically insulating.[14] Therefore, to make graphene oxide nanosheets electrically active again, graphene oxide nanosheets must be reduced in order to remove the oxygen containing moieties and thus, to restore the graphitic structure. To date, the reduction of graphene oxide nanosheets is performed by pyrolysis at high temperatures[15] or by chemical treatment using hydrazine, sodium borohydride, or hydroquinone as reducers.[13,14]

In practice, the reduction of graphene oxide nanosheets form GNSs results in a decrease in their hydrophilicity, which eventually leads to their irreversible agglomeration and precipitation.[14] Thus, the hydrophobic character of GNSs can affect the dispersion of GNSs in a polymer matrix during *in situ* polymerization in aqueous solution. The better the homogeneity is, the better the properties of the composites

*Corresponding Author. E-mail: dsohn@hanyang.ac.kr

such as the electrical conductivity.[16] Studies on the preparation of stable aqueous dispersions of GNSs in the presence of anionic polymers have been reported.[17,18] So far, polymer nanocomposites based on PANi and pristine graphite,[19] graphite oxide[20,21] as well as GNSs[22] have been investigated. However, to our knowledge, the modification of GNSs, especially non-covalent modification, for applications in PANi based nanocomposite is still limited.

In this paper, polymer nanocomposites based on polyaniline and water soluble GNS were prepared and characterized. The water soluble GNSs were prepared by reduction of graphene oxide nanosheets with hydrazine and simultaneous modification with poly(sodium 4-styrenesulfonate) (PSS). The polymer nanocomposites were prepared by *in situ* oxidative polymerization of aniline monomer with PSS-GNS in an acidic aqueous solution. The structure and properties of the polymer nanocomposites are investigated and discussed.

Experimental

Materials. Aniline (99.5%), nitric acid (70%), potassium chlorate (99%), poly(sodium 4-styrenesulfonate)(PSS) (99%), hydrazine hydrate (N_2H_4:50-60%) and natural graphite flakes with average diameter of 400 μm were purchased from Aldrich. Ammonium persulfate (APS) (98%), hydrochloric acid (37%) and sulfuric acid (97%) were purchased from SAMCHUN Chemical. All chemicals were used as received.

Preparation and Modification of Graphene Nanosheets. Graphene oxide nanosheets were prepared according to the Staudenmaier method.[23] A reaction flask was charged with 10 mL H_2SO_4 and 5 mL HNO_3 and cooled by immersion in an ice bath. After stirring and cooling for 15 min, 0.5 g natural graphite flakes were added under vigorous stirring. Then, 6 g $KClO_3$ was slowly added to the reaction mixture and allowed to stir for 5 days. On completion of the reaction, the mixture was filtered and washed several times with a 5% solution of HCl to remove all sulfate ions, followed by washing with distilled water. The acid intercalated graphite flakes were dried at 100 °C for 24 h. The thermal treatment was performed in a furnace at a temperature of 1,050 °C for 40 sec. Afterward, the worm-like expanded graphite flakes were sonicated in an ethanol solution (ethanol : water = 2 : 1, v/v) for 10 h. The graphene oxide nanosheets were obtained after filtering and washing the products with distilled water and drying in a vacuum oven for 24 h.

To prepare water soluble GNSs, graphene oxide nanosheets were reduced by hydrazine and simultaneously modified with PSS according to the procedure suggested by Stankovich[17] *et al*.. Graphene oxide nanosheets (0.05 g) and 0.5 g PPS were added to a reaction flask containing 50 mL distilled water. After sonication for 1 h, 3 mL hydrazine hydrate was added to the mixture. The reaction mixture was stirred at a temperature of 100 °C under a water-cooled condenser for 24 h. PSS modified GNSs were filtered and washed with distilled water and ethanol and dried in vacuum oven at room temperature for 24 h.

Synthesis of PANi/PSS-GNS Nanocomposites. PANi/PSS-GNS nanocomposites were prepared by *in situ* polymerization of aniline monomer in the presence of PSS-GNS. In a typical experiment, PSS-GNSs were dispersed in 5 mL ethanol solution (ethanol : water = 3 : 2, v/v) and then sonicated for 1 h. This suspension was added to a reaction flask containing 25 mL 1 M HCl solution and 0.9313 g aniline. The mixture was stirred for 30 min and then 25 mL 1 M HCl solution containing 1.14 g APS was added dropwise to the mixture. The reaction mixture was kept under static conditions for 6 h at 0-5 °C. After the reaction, the product was filtered, washed with distilled water followed by ethanol and dried in a vacuum oven at 50 °C for 6 h.

Characterization. The surface topological measurement of GNS was performed under ambient conditions with atomic force microscopy (AFM) (XE-100 microscopy, Park System Corp). The morphology of PANi/PSS-GNS nanocomposites was observed by scanning electron microscopy (SEM) (JEOL, JSM-840A). XRD experiments were performed on a Bruker-AXS D8 instrument with Cu Kα radiation (λ = 1.5406 Å) at 40 kV and 40 mA. ATR-FTIR spectra were collected on a FTIR spectrometer equipped with a microscope (SensIR Technologies, Danbury, CT). UV-vis absorption spectra of the samples were recorded on an Agilent 8453 instrument at ambient temperature. TGA experiments were performed on a TGA-SDT 2960 instrument in the temperature range from 25 to 800 °C with a heating rate of 10 °C/min in a N_2 atmosphere. Oxygen content in the sample was determined using an elemental analyzer (Flash EA 1112 series/CE Instrument). For the characterization of the electrical conductivity of the materials, the PANi/GNS-PSS composite powders were pressed into pellets at high pressure (7.5 ton/cm^2, 30 sec) with a thickness of around 300 μm and diameter of 13 mm. In the case of graphene oxide nanosheets, GNS and PSS-GNS, a film of these materials on a glass substrate was prepared by a dip-coating method in ethanol solution (0.001 g/mL). The thickness of the film was about 2.1 μm. The electrical conductivity of the samples was measured using the four probe method on a CMT-Series, Jandel instrument at room temperature.

Results and Discussion

Characterization of PSS Modified GNS. For the preparation of graphene oxide nanosheets, a thickness in the range of a single graphene layer is desirable. Figure 1 shows AFM images of graphene oxide nanosheets on a Si-wafer after the exfoliation process. It is clear that the obtained graphene oxide nanosheet has a thickness of around 3~4 nm. Assuming that the thickness of a single graphene layer

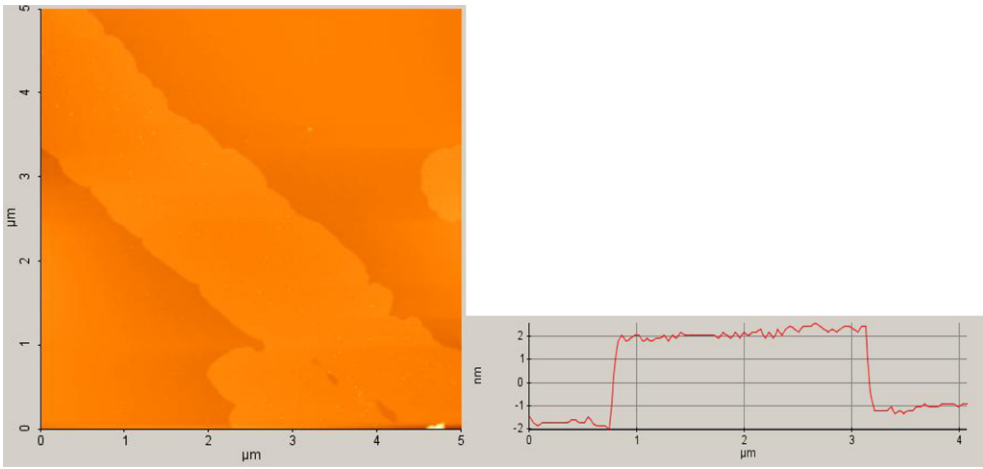

Figure 1. A non-contact mode AFM image of graphene oxide nanosheets with height profile.

is equal to the interlayer separation in graphite,[17,24] 0.34 nm, the obtained graphene oxide nanosheets likely consist of about 10 graphene layers.

The preparation and modification of GNS was performed by reduction of graphene oxide nanosheets with hydrazine in aqueous solution and simultaneously modification by non-covalent coating with PSS molecules through the π-π stacking of aromatic molecules on the graphene plane.[25,26] To prove the existence of non-covalent interactions between GNS and PSS, aqueous solutions of GNS, PSS and PSS-GNS (0.001 g/5 mL) were prepared and filtered through a PTFE syringe membrane filter (0.45 µm pore size) and subsequently the UV-vis spectra were recorded (Figure 2).

PPS has two absorption peaks at 225 and 261 nm and GNS has no absorption peaks. After modification with PSS, two absorption peaks appear at 228 and 282 nm for PSS-GNS. This result clearly indicates that the PSS molecules are successfully attached on the surface of GNS. The effect of coating GNS by PSS molecules is evident by the stability

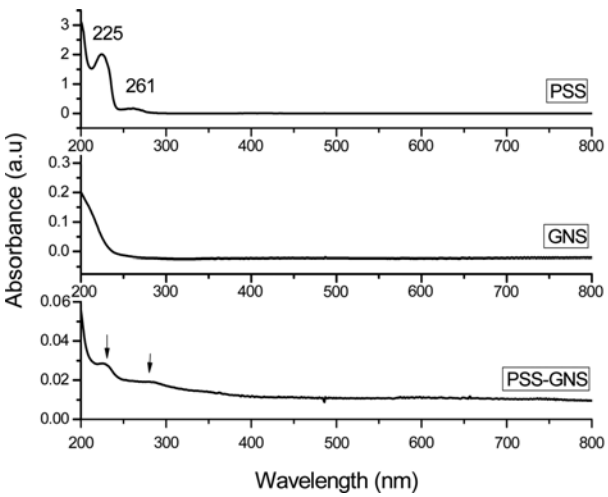

Figure 2. Comparison of UV-vis spectra of GNS, PSS, and GNS modified with PSS.

Figure 3. Images of water dispersion (0.001 g/mL) of GNS and PSS-GNS after 1 month.

of the dispersion of GNS with and without PSS in aqueous solution for long times (about 1 month) (Figure 3).

Meanwhile, the dispersion of PSS-GNS in water, prepared by precipitating GNS in an aqueous solution, is still stable. According to Stankovich[17] *et al*., the surface of GNS was partially covered with polymer due to defects in the graphitic structure of GNS.

Analysis of the Microstructure and Morphology of PANi/PSS-GNS Nanocomposite. Figure 4 shows the XRD patterns of pristine graphite, PSS-GNS, PANi and PANi/PSS-GNS (5%) nanocomposites. Pristine graphite has a very strong sharp peak at 26.6°, corresponding to the d-spacing between single graphene layers (0.34 nm). For PSS-GNS, the strong sharp peak of pristine graphite disappeared and a broad peak was observed at 23.4°. This XRD pattern of PSS-GNS is typical for disordered (turbostatic) graphitic platelets reported by Dresslhaus[27] *et al*.. This result confirms the successful exfoliation of pristine graphite into graphene nanosheets. The as-prepared PANi has a semi-crystalline structure. The main peaks in the PANi XRD pattern are located at 8.9, 15.0, 20.7, and 25.2°, which correspond to the

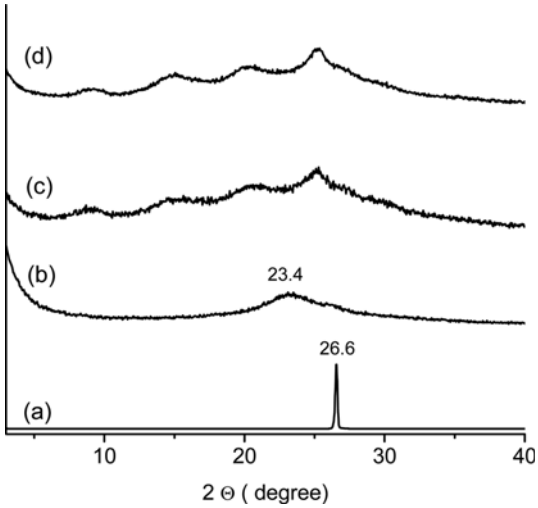

Figure 4. Comparison of XRD patterns of (a) pristine graphite, (b) PSS-GNS, (c) PANi, and (d) PANi/PSS-GNS (5%) nanocomposites.

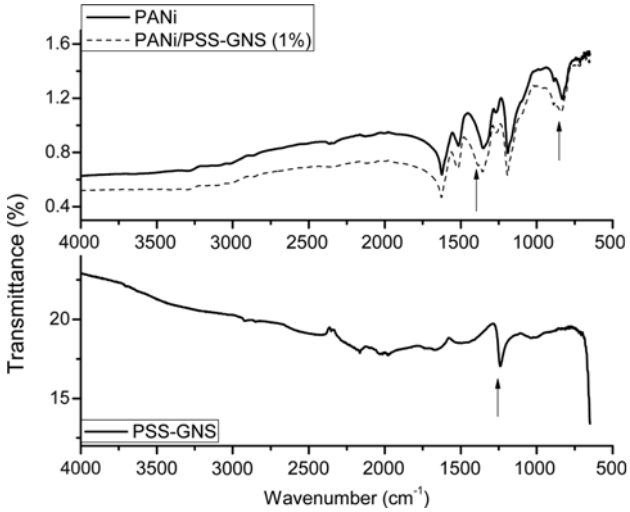

Figure 6. ATR-FTIR spectra of PANi, PANi/PSS-GNS (1%) nanocomposite and PSSGNS.

(001), (010), (100), and (110) reflection crystal planes. This crystal structure of PANi is of the ES-I crystalline type reported by Pouget[28] *et al.*. For the PANi/PSS-GNS nanocomposite, it is interesting to see that the reflection peak of PSS-GNS at 23.4° totally disappeared and the XRD pattern of the nanocomposite is almost identical to PANi. This result indicates that by *in situ* polymerization of aniline monomer with PSS-GNS, the disordered structure of PSS-GNS was fully destroyed and PSS-GNS was exfoliated in the PANi matrix.

Figure 5 shows the SEM-images of PANi and PANi/PSS-GNS (1%) nanocomposites. It is clear that PANi, that a rod-like morphology was formed, with diameters from 100 to 170 nm. In the polymer nanocomposite, the rod-like morphology of the neat PANi is less visible. Rather, a particle-like morphology was formed, and PANi is partly attached onto the surface of PSS-GNS. It is well known that, when HCl is used as a dopant, the aniline monomer was absorbed onto the surface of GNS through electrostatic attraction by the formation of weak charge-transfer complexes between aniline monomer and the graphitic structure of GNSs. As result of the absorption process, GNSs were coated by PANi particles by the *in situ* polymerization of aniline monomer

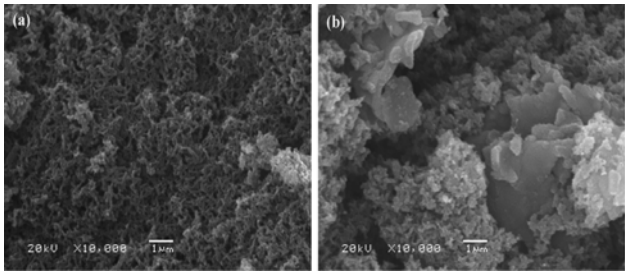

Figure 5. SEM-images of (a) PANi and (b) PANi/PSS-GNS (1%) nanocomposites.

in the presence of GNS.[21] In our case, GNSs were partly covered by PSS molecules. Thus, the PSS molecules prevented the absorption of aniline monomer onto the surface of GNSs, resulting in partial attachment of PANi on the surface of PSS-GNSs. Based on SEM images, it is clear to see that PSS-GNSs can exist as a single PSS-GNS or as a stack of several PSS-GNSs in a PANi matrix.

Analysis of ATR-FTIR Spectra of PANi/PSS-GNS Nanocomposites. Figure 6 shows the typical ATR-FTIR spectra of PSS-GNS, PANi, and PANi/PSS-GNS (1%) nanocomposites. For PSS-GNS, the ATR-FTIR spectrum is dominated by a sharp peak at 1242 cm^{-1}, which is assigned to C-O-C stretching.[18] A less intense, broad peak at 1470 and 1665 cm^{-1} can be assigned to the C-OH and C=O in carboxylic and carbonyl moieties.[18] This result indicates that a large numbers of epoxy groups were formed on the graphitic plane by the preparation of GNSs with chemical oxidation of graphite, PSS,[29] with characteristic peaks of S=O stretching at 1050 and 1030 cm^{-1} and S-O at 895 cm^{-1} appear as a weak broad peak at about at 1050 cm^{-1}. PANi[30] has the characteristic peaks of quinoid ring stretching at 1630 cm^{-1}, benzenoid ring stretching at 1512 cm^{-1}, C-N stretching at 1347 cm^{-1}, N=Q=N, (Q is a quinoid) at 1191 cm^{-1} and C-H out of plane bending at 825 cm^{-1}. For the PANi/PSS-GNS nanocomposite, the ATR-FTIR spectrum is almost the same as the PANi spectrum, except a new peak, which appears at 1376 cm^{-1} and the C-H bending peak shifted to a higher wavenumber (846 cm^{-1}). Both PANi and PSS-GNS do not have any characteristic peaks at 1376 cm^{-1}, and the peak of C-O-C stretching of PSS-GNS is missing. Therefore, it is believed that the new peak at 1376 cm^{-1} in the PANi/PSS-GNS nanocomposite is a result of shifting of the C-O-C bonding peak to a higher wavenumber. According to Levitt[31] *et al.*, the aromatic ring can act as a hydrogen bond acceptor.

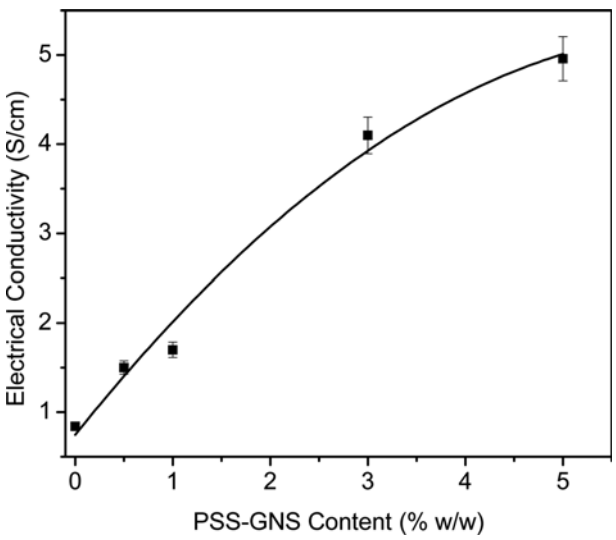

Figure 7. Dependence of the electrical conductivity of PANi/PSS-GNS nanocomposite on the PSS-GNS content.

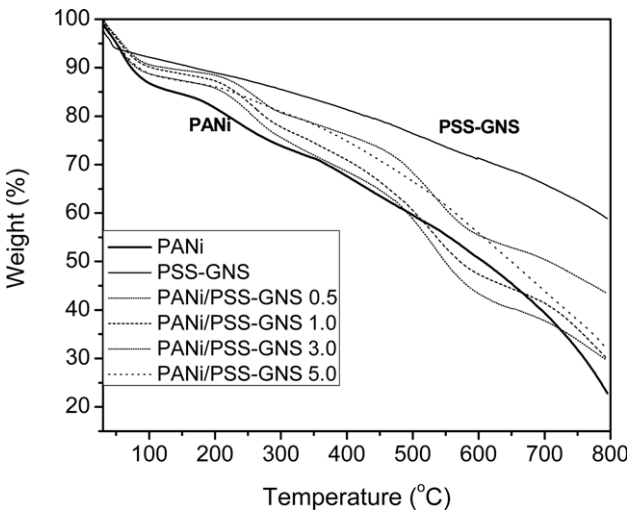

Figure 8. TGA thermograms of PANi, PSS-GNS, and PANi/PSS-GNS nanocomposite prepared with different PSS-GNS contents (the numbers in the figure indicate the content of PSS-GNS in the nanocomposite in %, w/w).

Thus, the shifting of both C-O-C bonding in PSS-GNSs and C-H bending in PANi to higher wavenumber in the nanocomposite clearly indicates that hydrogen bonding exists between oxygen atoms in the epoxy group of PSS-GNSs and hydrogen atoms of aromatic rings of PANi.

Electrical Conductivity. Electrical conductivity is one of the most important properties of composites for applications in optoelectronic devices. The electrical conductivity of PANi/PSS-GNS nanocomposites as a function of PSS-GNS content is shown in Figure 7. It is clear to see that, with increasing PSS-GNS content, the electrical conductivity of PANi/PSS-GNS nanocomposites strongly increases from 0.84 S/cm for neat PANi to 4.96 S/cm for the PANi/PSS-GNS (5%) nanocomposite. Errors in the electrical conductivity values were about 3%. Graphene oxide nanosheets containing oxygen moieties have very low electrical conductivity, about 1.7×10^{-7} S/cm. According to the results of elemental analysis, the oxygen content in graphene oxide nanosheets decreases from 8.88% (± 0.17) to 7.18% (± 0.02) after reduction with hydrazine in aqueous solution. The removal of the oxygen moieties resulted in an increasing in electrical conductivity of GNS to 0.3 S/cm, about 6 orders of magnitude better than the electrical conductivity of graphene oxide nanosheets. This electrical conductivity of GNS is close to the electrical conductivity of reduced graphene oxide (0.24 S/cm) reported by Stankovich[14] *et al.*. After modification of GNS with PSS, the electrical conductivity of PSS-GNS slightly decreases to 0.14 S/cm. According to Du[21] *et al.* in PANi/PSS-GNS nanocomposites, PSS-GNS could serve as an electrically conductive bridge in the PANi matrix, and thus increases the electrical conductivity of the polymer nanocomposite. At high PSS-GNS content (more than 3%), a strong increase in the electrical conductivity of PANi/PSS-GNS nanocomposite was observed. It is possible that at high PSS-GNS content, a conductive network of PSS-GNS formed, resulting in the strong increase in electrical conductivity in the nanocomposite.

Thermal Stability. Figure 8 shows the TGA thermograms of PANi, PSS-GNS and PANi/PSS-GNS nanocomposites with different PSS-GNS content. PSS-GNSs do not show any significant weight loss in the temperature range except weight loss below 100 °C from the moisture absorbed in the sample. Three weight loss steps are observed in PANi. The first weight loss below 100 °C is from moisture absorbed in the sample. The second weight loss from 170 to 270 °C is due to the weight loss of the dopant. Finally, the polymer begins thermally to degrade at a temperature of 360 °C. The thermal behavior of the doped PANi is the same as that reported by Bhadra[32] *et al.*. For the PANi/PSS-GNS nanocomposite, three weight loss steps are also observed. The first two weight loss steps are identical to neat PANi. The main difference lies in the weight loss step of the polymer. In the nanocomposite the degradation temperature of PANi is significantly increased from 360 °C in neat PANi to 460 °C. Further, the thermal behavior of PANi/PSS-GNS nanocomposite with 5% PSS-GNS is similar to that of pristine PSS-GNS. This result clearly indicates that the thermal stability of PANi is strongly improved by incorporation of PSS-GNS.

Conclusions

In this paper, PANi/PSS-GNS nanocomposites were prepared and characterized. By the *in situ* polymerization process, the disordered structure of PSS-GNS was fully destroyed and both single PSS-GNS and stacked PSS-GNS were dispersed in a PANi matrix. Interaction occurs between PANi

and PSS-GNS through hydrogen bonding between the epoxy group in PSS-GNS and the aromatic ring of PANi. By the incorporation of PSS-GNS, the electrical conductivity of PANi strongly increased from 0.84 S/cm in neat PANi to 4.96 S/cm in PANi/ PSS-GNS nanocomposite with 5% PSS-GNS. The thermal stability of PANi was also significantly improved from 360 °C in neat PANi to 460 °C in PANi/PSS-GNS nanocomposites. These intriguing features of the nanocomposites make them promising materials for applications in electronic devices.

Acknowledgements. This work was supported by a research fund from Hanyang University.

References

(1) S. Bhadra, D. Khastgir, N. Singha, and H. J. Lee, *Prog. Polym. Sci.*, **34**, 783 (2009).
(2) J. Unsworth, B. A. Lunn, D. C. Innis, Z. Jin, A. Kaynak, and N. G. Boot, *J. Intell. Mater. Syst. Struct.*, **3**, 380 (1992).
(3) N. Gospodinova and C. Terlemezgan, *Prog. Polym. Sci.*, **23**, 1443 (1980).
(4) N. Kuramoto and A. Tomita, *Synth. Met.*, **88**, 147 (1997).
(5) Y. Leroux, E. Eang, C. Fave, G. Trippe, and J. C. Lacroix, *Electro. Comm.*, **9**, 1258 (2007).
(6) T. K. Sarma, D. Chowdhury, A. Paul, and A. Chattopadhyay, *Chem. Comm.*, **9**, 1048 (2002).
(7) K. Gupta, P. C. Jana, and A. K. Meikap, *Synth. Met.*, **160**, 1566 (2010).
(8) Y. Qiao, C. M. Li, S.-J. Bao, and Q.-L. Bao, *J. Power Source*, **170**, 78 (2007).
(9) G. Chen, W. Weng, D. Wu, C. Wu, J. Lu, P. Wang, and X. Chen, *Carbon*, **42**, 753 (2004).
(10) A. K. Geim and K. S. Novoselov, *Nature Materials*, **6**, 183 (2007).
(11) C. D. Hodgman, *Handbook of Chemistry and Physics,* 42nd Ed., Chemical Rubber Publishing, 1960.
(12) S. Park and R. S. Ruoff, *Nature Nanotechnology*, DOI 10.1038/WNANO.2059.58.
(13) A. B. Bourlinos, D. Gournis, D. Petridis, T. Szabo, A. Szeri, and I. Dekany, *Langmuir*, **19**, 6050 (2003).
(14) S. Stankovich, D. A. Dikin, R. D. Piner, K. A. Kohlhaas, A. Kleinhammers, Y. Jia, Y. Wu, S. B. T. Nguyen, and R. S. Ruoff, *Carbon*, **45**, 1558 (2007).
(15) M. Hirata, T. Gotou, and M. Ohba, *Carbon*, **43**, 505 (2005).
(16) J. Yong, H. M. Jeong, and B. K. Kim, *Macromol. Res.*, **17**, 626 (2009).
(17) S. Stankovich, R. D. Piner, X. Chen, N. Wu, S. B. T. Nguyen, and R. S. Ruoff, *J. Mater. Chem.*, **16**, 155 (2006).
(18) Y. Si and E. T. Samulski, *Nano Letter*, **8**, 1679 (2008).
(19) S. E. Bourdo and T. Viswanathan, *Carbon*, **43**, 2983 (2005).
(20) H. Wang, Q. Hao, X. Yang, L. Lu, and X. Wang, *Electro. Comm.*, **11**, 1158 (2009).
(21) X. S. Du, M. Xiao, and Y. Z. Meng, *Eur. Polym. Sci.*, **40**, 1489 (2004).
(22) J. Yan, T. Wei, B. Shao, Z. Fan, W. Qian, M. Zhang, and F. Wei, *Carbon*, **48**, 487 (2010).
(23) L. Staudenmaier, *Ber. Deutsch. Chem. Ges.*, **31**, 1481 (1898).
(24) W. L. Zhang, B. J. Park, and H. J. Choi, *Chem. Comm.*, **46**, 5596 (2010).
(25) Y. Xu, H. Bai, G. Lu, C. Li, and G. Shi, *J. Am. Chem. Soc.*, **130**, 5856 (2008).
(26) K. P. Loh, Q. Bao, P. K. Ang, and J. Yang, *J. Mater. Chem.*, **20**, 2277 (2010).
(27) M. S. Dresselhaus, *Supercarbon: Synthesis, Properties and Application*, S. Yoshimura and R. P. H. Chang, Eds., Springer, New York, 1988, vol. 33, p.9.
(28) J. P. Pouget, M. E. Jozefowicz, A. J. Epstein, X. Tang, and A. G. MacDiarmid, *Macromolecules*, **24**, 779 (1991).
(29) X. Zhao, N. Song, X. Chen, X. Fan, and Q. Zhou, *J. Mater. Chem.*, **16**, 4619 (2006).
(30) K. G. Neoh, E. Tang, and K. L. Tan, *Synth. Met.*, **60**, 13 (1993).
(31) M. Levitt and M. F. Perutz, *J. Mol. Biol.*, **201**, 751 (1988).
(32) S. Bhadra, N. K. Singha, and D. Khastgir, *J. Appl. Polym. Sci.*, **104**, 1900 (2007).